

If we suppose that the outfielder is standing where the ball will eventually land, at $x = 2$, we can get a sense of the way the angle of the outfielder's line of sight with the ball increases by plotting a line of sight from early in the ball's trajectory. The following line of code creates a line segment for the plot we drew in [Listing 1-2](#), and it is meant to be run in the same Python session. This line segment represents the line between the outfielder's eyes and the ball after the ball has traveled 0.1 meters horizontally.

```
xs2 = [0.1, 2]
ys2 = [ball_trajectory(0.1), 0]
```

We can plot this line of sight along with other lines of sight to see how the angle continues to increase over the course of the ball's trajectory. The following lines of code add more line segments to the same plot we drew in [Listing 1-2](#). These line segments represent the line between the outfielder's eyes and the ball at two more points in the ball's journey: the points when the ball has traveled 0.1, 0.2, and 0.3 meters horizontally. After creating all of these line segments, we will plot them all together.

```
xs3 = [0.2, 2]
ys3 = [ball_trajectory(0.2), 0]
xs4 = [0.3, 2]
ys4 = [ball_trajectory(0.3), 0]
plt.title('The Trajectory of a Thrown Ball - with Lines of Sight')
plt.xlabel('Horizontal Position of Ball')
plt.ylabel('Vertical Position of Ball')
plt.plot(xs, ys, xs2, ys2, xs3, ys3, xs4, ys4)
plt.show()
```

The resulting plot shows several lines of sight that form continuously increasing angles with the ground ([Figure 1-2](#)).